Imaging Pipeline: Continuity and Page Classification

How pages become documents, what happens to the page-classification model's output, and how a page's Extracted type becomes its Final type. Reflects the code on main as of 2026-10-23 (commit dae7328). Thresholds and points are read from the canon files (*keyword-canon/continuity_canon.json*, *page_arbitration.json*, *page_canon_v24.json*), which can be edited without a code change.

Where each step runs

```
page_classify      ->  Extracted page type (page_classification)
continuity         ->  documents (page_continuity_results, continuity.json)
encounter          ->  encounter type per document
imaging_final      ->  Final page type, specialty, DOS, encounter (imaging_final)
```

Continuity never reads a Final value, so a Final value cannot feed back into how the pages were grouped.

1. Continuity logic: which pages form one document

Continuity walks the chart in page order and scores every **gap** between a page and the one before it. A positive score is a **break** (a new document starts), a negative score **joins** the page to the open document. Blank and duplicate pages are skipped; junk pages are scored only as runs (1.5).

1.1 What is read about each page

• OCR lines with positions (Final OCR 2, else Final OCR 1, else Tesseract), and section headings.

• Page type: its v24 L1 family and every name on its path, with the classifier's confidence and whether a canon title heads the page.

• DOS and how it was found (label = 5 points of trust, edge keyword = 3, keyless = 2).

• Signatures (anywhere / in the closing lines), printed "page N of M", visit/encounter numbers.

• Blank/junk verdict, and whether the page is **Visual** (page-tag visual, fewer than 20 words).

1.2 The points

Points are 1, 2, 3 or 5 only. 5 is decisive. Each signal sits on one side of the gap: **ending** (the earlier page alone), **opening** (the later page alone) or **pair** (the two compared). + pushes toward a break, - toward a join. "Scaled" signals are multiplied by a 0-1 strength.

Signal	Side	Points	Fires when
different_dos / same_dos	pair	+5 / -5 (x trust)	The later page's DOS differs from / equals the open document's. Decisive only when both dates were read from a label (trust 5).
pagination_follows	pair	-5, decisive	Printed page numbers run on (page 2 of 3 after page 1 of 3).
signed_in_tail / signed_anywhere	ending	+3 / +2	The earlier page is signed in its closing lines / anywhere.
last_numbered_page	ending	+3	The earlier page prints "page N of N".
has_title	opening	+3	A canon title heads the later page.
visit_heading	opening	+3 (and forced break, 1.4)	Reason for Visit / Appointment, Chief Complaint, CC, HPI, Visit Information at the start of a line in the top third.
page_type_change	pair	+3 scaled	(1 - same-type probability) x the classifier's confidence.
junk_run	pair	+3 / +5 scaled	Edge of a run of 2 / 3+ junk pages, x mean junk confidence.
says_continued	ending	-3	"continued", "cont'd", "see next page" at the end.
sections_forward	pair	-3 scaled	Section headings move forward in note order (HPI -> Exam -> Plan).
same_visit_number	pair	-3	Both pages carry the same visit / FIN / CSN number.
page_fill	ending	+2 scaled	Where the earlier page's text stops: a short page leans break, a full page leans join.
same_header_line / same_footer_line	pair	-2 / -1	A repeated header / footer line.

Signal	Side	Points	Fires when
wording_char / wording_word	pair	-2 / -1 scaled	The end of one page reads on into the top of the next (character / word similarity).
cut_mid_sentence / starts_mid_thought	ending / opening	-1 / -1	Tie-breakers beside a wording signal.
same_page_type	pair	-1	Both pages carry the same full type path.
same_junk_kind / blank_between	pair	-2 / +1	Two junk pages of one kind / a blank page between.

1.3 Deciding a gap

• **Score > 0**: break. **Score < 0**: joined. **Score = 0**: joined and flagged (decided_by tie).

• **A 5-point signal settles the gap** when no other 5 opposes it (decided_by decisive, high confidence).

• **Two decisive signals disagree** (page numbers run on, dates differ): the different date wins, medium confidence, flagged (decisive_conflict).

• Confidence: P(break) = 1 / (1 + e^(-score / 2)), confidence = |2P - 1|; x 0.6 when the ending, opening and pair sub-scores point opposite ways; at least 0.9 for a decisive gap. Level: |score| >= 5 high, >= 2 medium, else low.

1.4 Forced rules (2026-10-23), applied after the score

Rule	Effect	Canon switch
Demographics boundary	A Demographics / Patient Demographics / Information / Registration page next to any other kind of page is always a break. Consecutive Demographics pages stay together; standing alone it takes DOS 2222-02-02.	demographics_break
Visit heading	A visit heading in the top band always starts a new document (a new Progress Note), unless the printed page N of M runs on from the page before.	visit_headings.decisive
Visual page	A Visual page that is not junk always joins the page before it. Applied last, so it wins.	visual_join.enabled

1.5 Junk runs

Two or more junk pages in a row are scored like content and broken off at both ends (junk_run). A run of three or more scoring >= 4.5 is decisive. The run becomes its own **junk group** ("New (Junk 4)" ... "Last (Junk 4)"), numbered with the documents; its DOS and encounter type read Skipped.

1.6 The final pass: whole documents

Step	What it does
Split	A document holding two or more different dates read from a label splits at each date change.
Merge (weak breaks only)	Up to 3 passes, only across a break with low confidence or score <= 2, never a decisive one, never across different dates: **same_date** (equal dates), **continued_across** (first ends "continued"), **same_type_untitled** (same type, neither titled), **orphan_page** (one undated page between two documents of one type), **junk_rejoin** (two junk groups of one kind).
Progress Note runs on	A document opening on a Progress Note absorbs the documents after it until a new labelled date, a signature on its last page, a decisive break (including the forced breaks above), or a junk group.

1.7 What it writes

• **page_continuity_results**: per content or junk page - document_seq, seq, position (New / Continue / Last / Standalone), score, confidence, decided_by, evidence, link_strength, start_confirmed, review_required.

• **staging/continuity.json** (version 2): the documents, then per page both gaps in plain words (Break / Same document, score, how sure, up to 4 reasons) and the Final values. Every score: **continuity_detail.json**.

• **imaging/<chart>_continuity.csv**: every page.

2. Page classification: what happens to the model's output

The page-axes model reads the page text and answers on five axes. Its answer is never used raw: each level passes a gate, and below the gate the **v24 keyword canon** decides. Code: *page_classify/axes.py* (gate) and *axes_model.py* (decode).

2.1 The raw model output (decode)

Field	What it holds
L1, L1_confidence, L1_top	The best family and the top 3 with probabilities.
L2, L3	The best L2 under the predicted L1 (masked to that L1's L2s), and the best L3 under that L2.
by_L1 (new)	For every L1 that has L2s, the best L2 (and L3 under it) the model gives **under that L1**. Used by Final when the document has another L1 (3.3).
specialty, specialty_top	Best specialty and the top 3.
codeability	The model's codeability - kept for reference only, never used.

2.2 The gate: who decides each level

Level	Model decides when	Otherwise
L1	Top L1 >= 0.40, ahead of the second by >= 0.10, and not the folded OTHER class (route = model).	Canon: the best L1 among all canon entries at keyword confidence >= 0.50. Nothing clears: Unknown, needs review.
L2	L1 came from the model and the model's L2 >= 0.50.	Canon: the best L2 under that L1 at >= 0.50. Not decided for Progress Note (its L2 is the content section).
L3	Never (the model's L3 is kept in the raw prediction only).	Canon: the best L3 under L1 > L2 at >= 0.50.
Specialty	Not Unknown/Others, >= 0.70, ahead of the second by >= 0.20, trained on >= 10 pages.	The specialty of the canon entry behind the decided path, if it has one.
Codeability	Never.	The tag of the canon page type behind the decided path. A context / source rule stays needs_review until Final (3.4).

Keyword confidence: a title hit 0.60, a body hit 0.30, +0.05 per point of margin over the runner-up, capped.

2.3 under_l1: the page's place under every other L1 (new)

For every other L1 with L2s (except Progress Note), the classifier records the L2 / L3 the page would take there, by the same gates: the model's by_L1 L2 at >= 0.50, else the canon's best L2 under that L1 at >= 0.50; L3 the canon's at >= 0.50. Stored in *page_classification.axes.under_l1*. This is what lets Final change a page's L1 without breaking the hierarchy.

2.4 Naming the Extracted page type

Levels decided	page_type	page_subtype	Example
L1 only	L1	**Others** (never L1 again)	Progress Note / Others
L1 + L2	L1	L2	Medication Record / Medication List
L1 + L2 + L3	L1 (L2)	L3	Lab Report/Result (Radiology) / CT Scan Report
Nothing	Unknown	Unknown	needs review

Blank, junk and duplicate pages are not classified: their sub-type is the junk label. A duplicate takes its original's page type.

2.5 What it writes

page_classification: page_type, page_subtype, model_type (L1 > L2 > L3), classification_category (codeable / non_codeable / discharge_summary), confidence (L1's), decided_by (axes:model/canon/...), needs_review, and one **axes** JSON with every level, its confidence and source, specialty, codeability rule, title_l1, under_l1 and the raw prediction.

3. From Extracted to Final

Extracted is the page on its own (2.4). **Final** is the page as part of its continuity document, written by the imaging_final stage (*arbitration.decide_final*). Every page of a document shares one L1; L2 / L3 always fit under it in the v24 hierarchy.

3.1 Step 1: the document agrees on one L1

Basis	When	Agreed L1
first_page	The document's first page has an L1 (not Unknown).	The first page's L1.
vote	The first page is Unknown.	The L1s weighted by page-type confidence, when one holds >= 60% of the weight (flagged for review).
no_agreement	No L1 reaches 60%.	None: each page keeps its own type, flagged for review.

3.2 Step 2: each page's Final type

Rule	When	Final page type / sub-type
agreed	The page's L1 is the agreed L1.	Kept as Extracted (an L1-only page reads L1 / Others).
agreed_first_page / agreed_vote	The page's L1 differs (page_type_source = continuation).	See 3.3. A vote, or a weak link to the document, flags the page for review.
no_agreement	The vote had no majority.	Kept as Extracted, flagged.
embedded_same_page	A page in no multi-page document, of an embedded family, headed by a Progress Note title.	Progress Note / its own L2 (else its family).
no_change	A page in no multi-page document.	Kept as Extracted.

3.3 Choosing L2 / L3 under the new L1

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Agreed L1 = Progress Note?
  page family embedded (Lab Report/Result, Medication Record, Immunization,
  Problem List, Order and Requisition)?  ->  Progress Note / page's own L2 (else family)
  otherwise                               ->  Progress Note / Others
Any other L1:
  under_l1[L1] has an L2 (model >= 0.50, else canon >= 0.50)?
      with L3  ->  "L1 (L2)" / L3
      no L3    ->  L1 / L2
  row classified before under_l1: own name is an L2 of that L1?  ->  L1 / that name
  nothing fits                                                    ->  L1 / Others
```

Example: a Medication List page inside a Progress Note document is **Progress Note / Medication List**; a Billing Statement there is **Progress Note / Others**; a Progress Note page inside a Discharge Summary document takes the Discharge Summary L2 it scored, else **Discharge Summary / Others**.

3.4 Codability

• A page that keeps its own L1 keeps its own canon codability.

• A page that takes the agreed L1, or whose canon rule is *context* (e.g. Clinical Summary), takes the **document's** codability: that of its first page of the agreed family.

• Inside a Progress Note, a page carrying Laboratory, Laboratory Data, Medication Record or Medication List **stays Non-Codable**; radiology does not.

3.5 Values carried across the document

Field	Final value
Specialty	The first page of the document that has a specialty; every page takes it and it never changes midway. Unknown when none has one.

Field	Final value
DOS	The first dated page's (the first page's when dated). A stand-alone non-visit document (demographics, registration, problem list, immunizations ...) takes 2222-02-02.
Encounter type	The document's, decided once: only for Progress Note / Discharge Summary documents (else Not Applicable); a Progress Note with no evidence defaults to Outpatient F2F at 0.3.
Junk pages	Keep their own values whatever their document number.

3.6 Worked examples

Document	Page	Extracted	Final
A	1 DISCHARGE SUMMARY, page 1 of 3	Discharge Summary / Others	Discharge Summary / Others (agreed)
A	2 untitled note, page 2 of 3	Progress Note / Others	Discharge Summary / its L2 there, else Others
A	3 lab table, page 3 of 3	Lab Report/Result (Laboratory) / Chemistry Report	Discharge Summary / Others
B	1 OFFICE VISIT	Progress Note / Others	Progress Note / Others
B	2 lab results	Lab Report/Result (Laboratory) / Chemistry Report	Progress Note / Laboratory, Non-Codable
B	3 medication list	Medication Record / Medication List	Progress Note / Medication List, Non-Codable
C	1 Unknown, 2-3 lab (0.90 / 0.80)	Unknown; Lab Report/Result x2	Lab Report/Result / Others on page 1, review (agreed_vote)

Re-run note. Pages classified before 2026-10-23 have no under_l1 and may still read L1 / L1. Re-run page classification and the stages after it (continuity, encounter, imaging_final) to get the Final values described here.